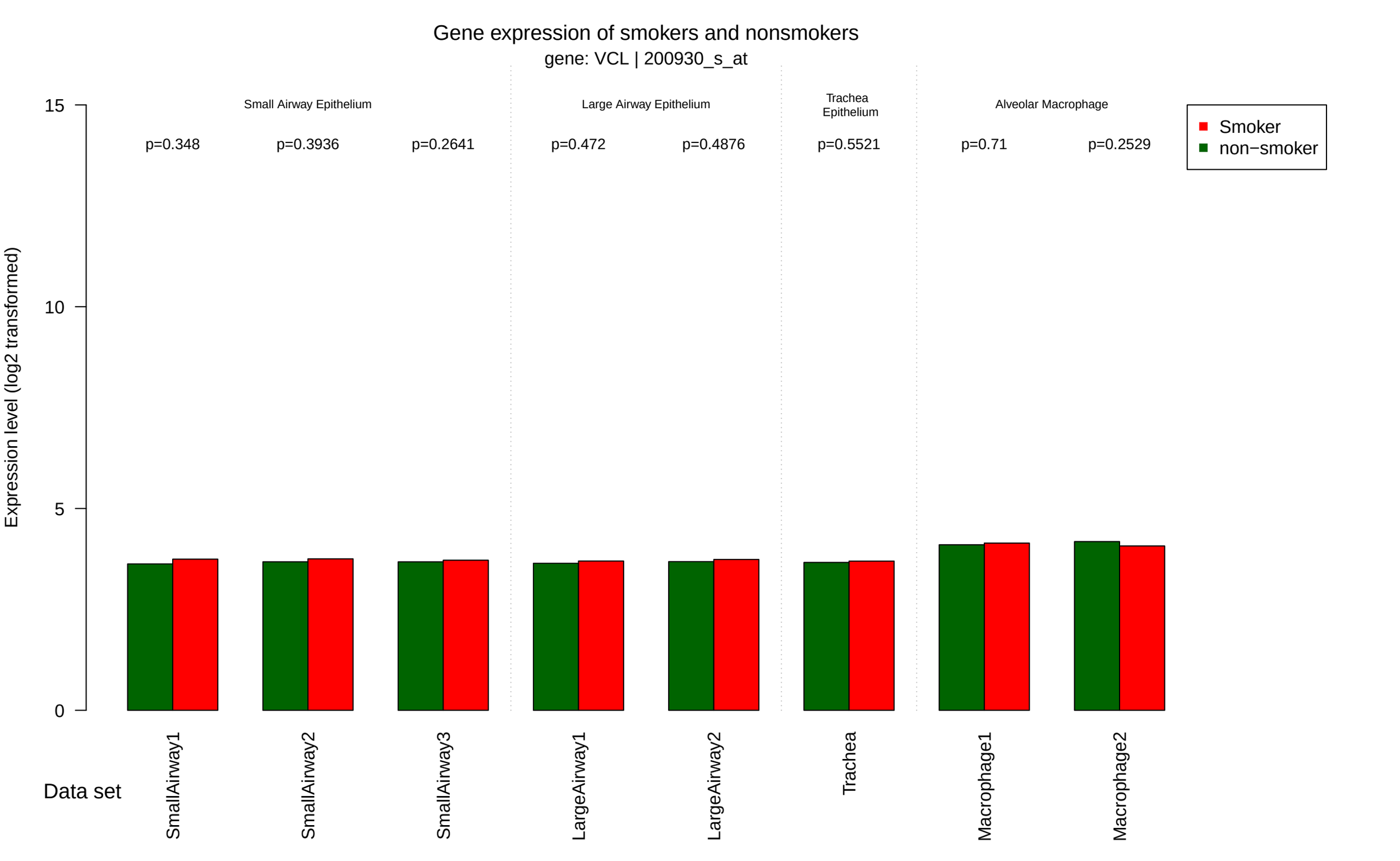
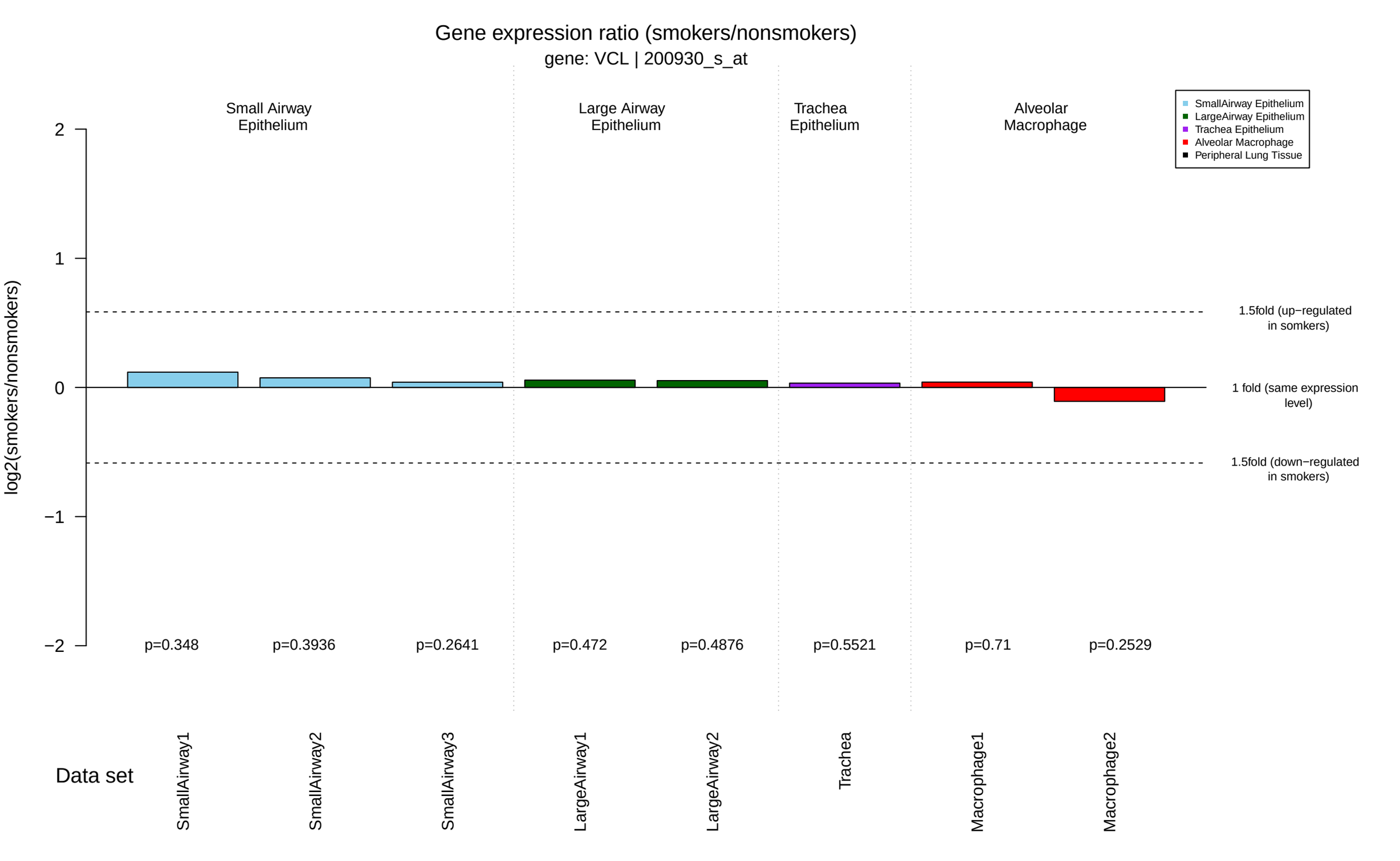
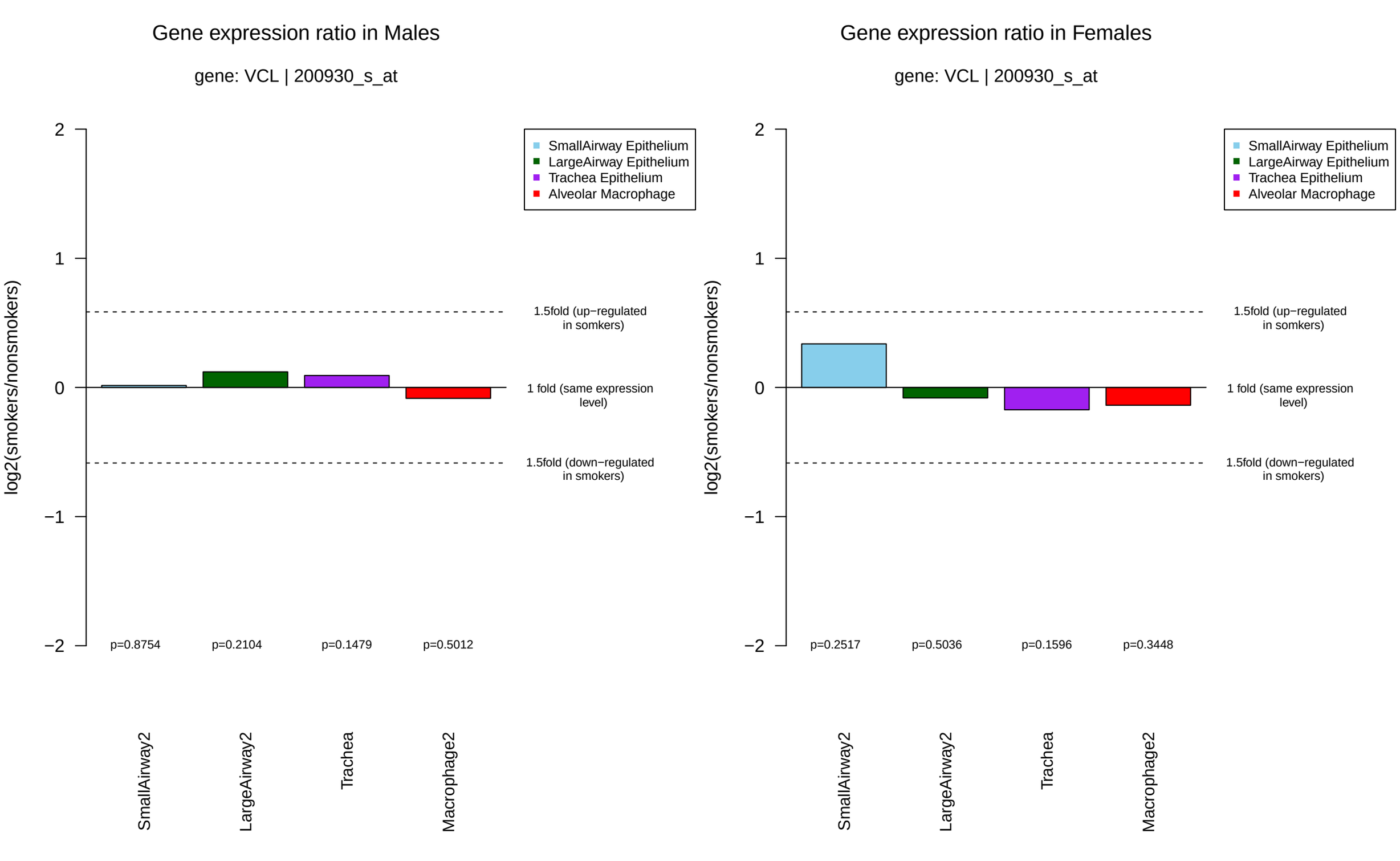
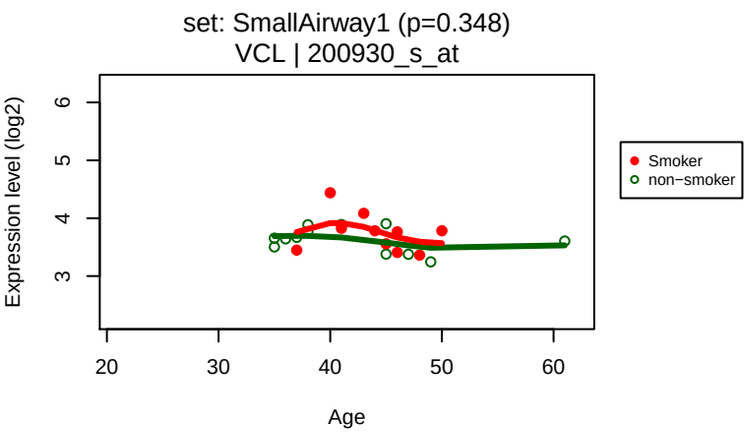
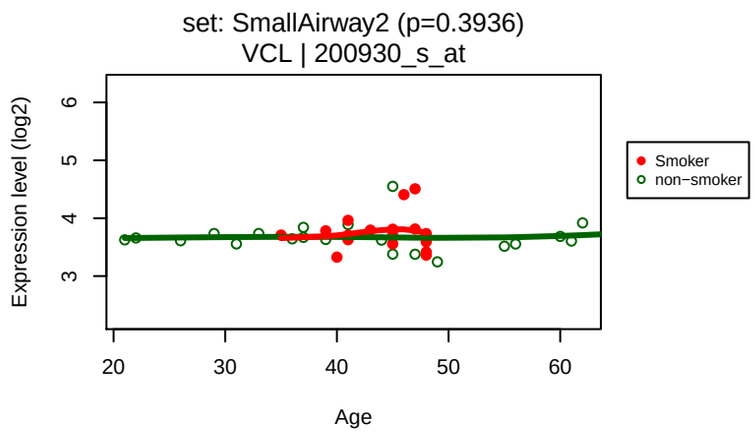
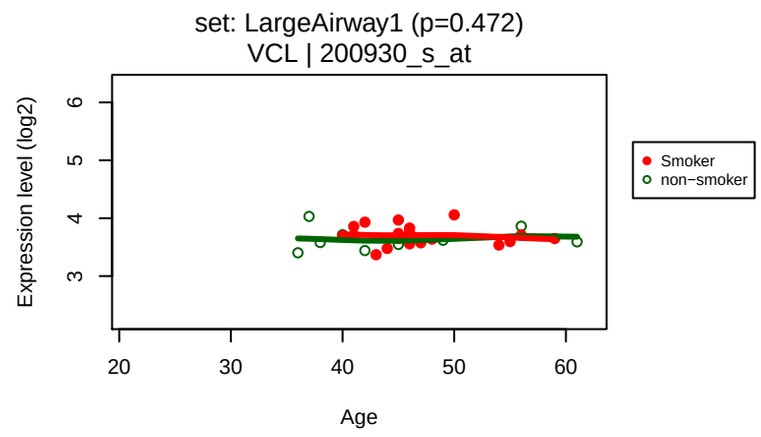

Gene expression of smokers and nonsmokers
gene: VCL | 200930_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.348
p=0.3936
p=0.2641
p=0.472
p=0.4876
p=0.5521
p=0.71
p=0.2529
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

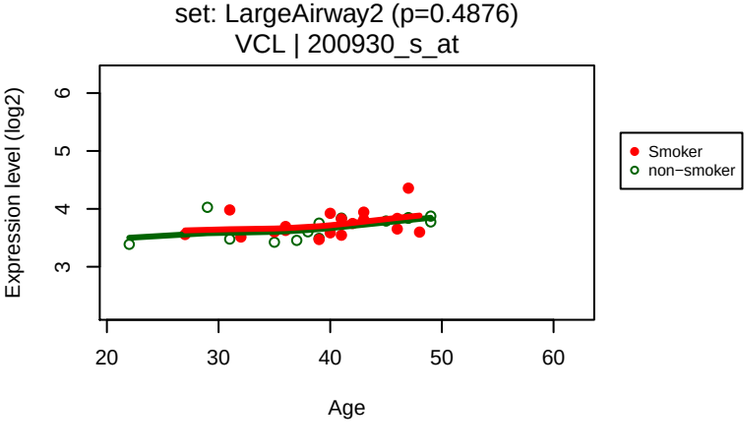

Gene expression ratio (smokers/nonsmokers)
gene: VCL | 200930_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.348
p=0.3936
p=0.2641
p=0.472
p=0.4876
p=0.5521
p=0.71
p=0.2529
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

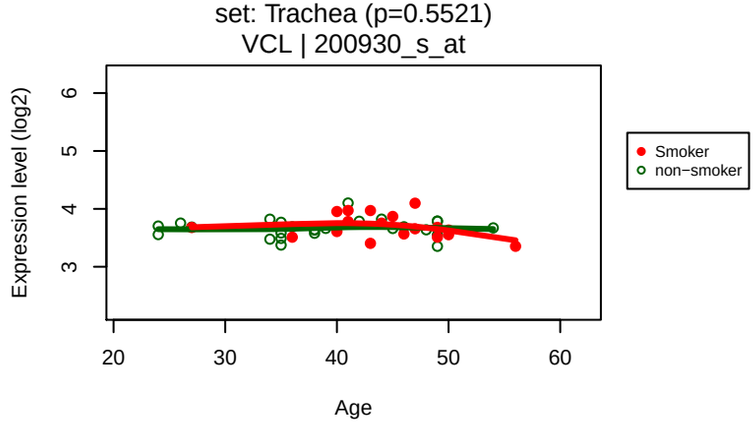

Gene expression ratio in Males
gene: VCL | 200930_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8754
p=0.2104
p=0.1479
p=0.5012
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VCL | 200930_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2517
p=0.5036
p=0.1596
p=0.3448
SmallAirway2
LargeAirway2
Trachea
Macrophage2

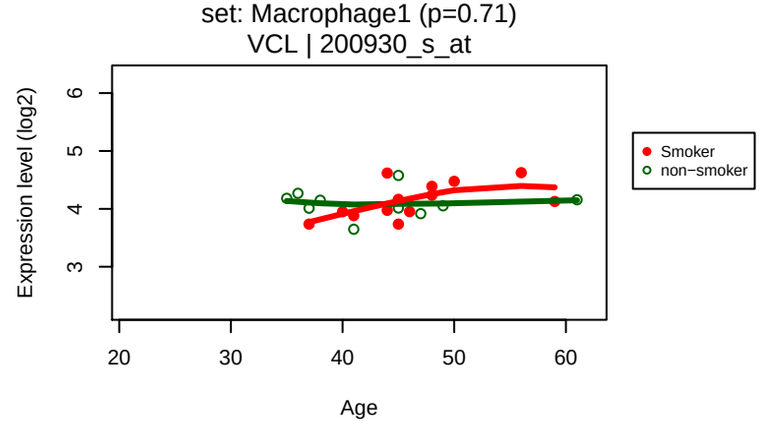

set: SmallAirway1 (p=0.348)
VCL | 200930_s_at
Expression level (log2)
Age
Smoker
non−smoker

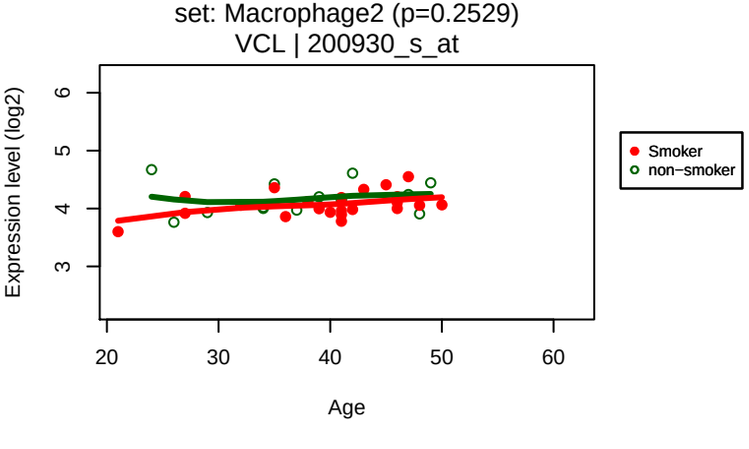

set: SmallAirway2 (p=0.3936)
VCL | 200930_s_at
Expression level (log2)
Age
Smoker
non−smoker

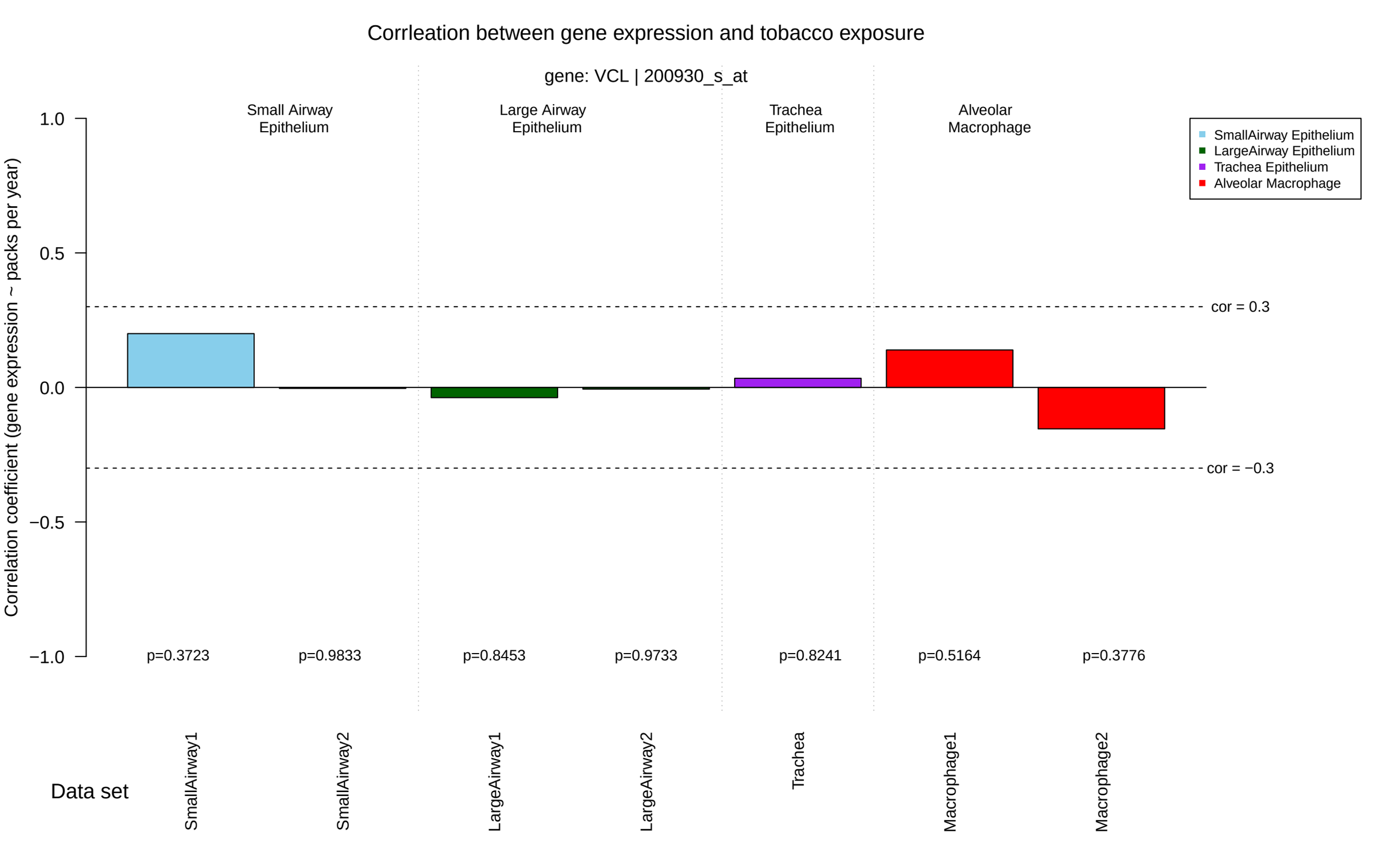

set: LargeAirway1 (p=0.472)
VCL | 200930_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4876)
VCL | 200930_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5521)
VCL | 200930_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.71)
VCL | 200930_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2529)
VCL | 200930_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VCL | 200930_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3723
p=0.9833
p=0.8453
p=0.9733
p=0.8241
p=0.5164
p=0.3776
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2